Lab 7.1: Error Debugging with AI – Systematic Approaches to Finding and Fixing Bugs

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Lab Objectives

• To identify and correct syntax, logic, and runtime errors in Python programs using AI tools.
• To understand common programming bugs and AI-assisted debugging suggestions.
• To evaluate how AI explains, detects, and fixes different types of coding errors.
• To build confidence in using AI for systematic debugging.

Lab Outcomes

• Detect and correct syntax, logic, and runtime errors.
• Understand AI explanations for bugs.
• Apply structured debugging strategies.
• Refactor buggy code safely and correctly.

Task 1: Syntax Error – Missing Parentheses in Print Statement

Buggy Code:

```
def greet():
    print "Hello, AI Debugging Lab!"
```

Observed Error:
SyntaxError: Missing parentheses in call to 'print'

AI Explanation:
Python 3 requires parentheses for the print() function.

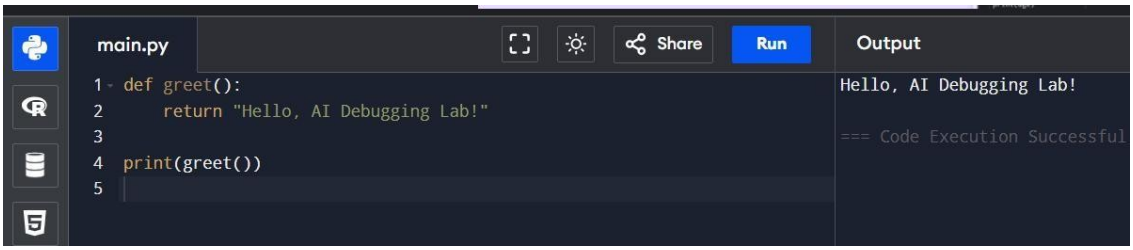
Corrected Code:

```
def greet():
    return "Hello, AI Debugging Lab!"

print(greet())
```

Assert Test Cases:
assert greet() == "Hello, AI Debugging Lab!"
assert isinstance(greet(), str)
assert greet().startswith("Hello")

Output:
Hello, AI Debugging Lab!



Task 2: Logic Error – Incorrect Condition in If Statement

Buggy Code:

```
def check_number(n):
    if n = 10:
        return "Ten"
```

AI Explanation:
= is assignment, == is comparison.

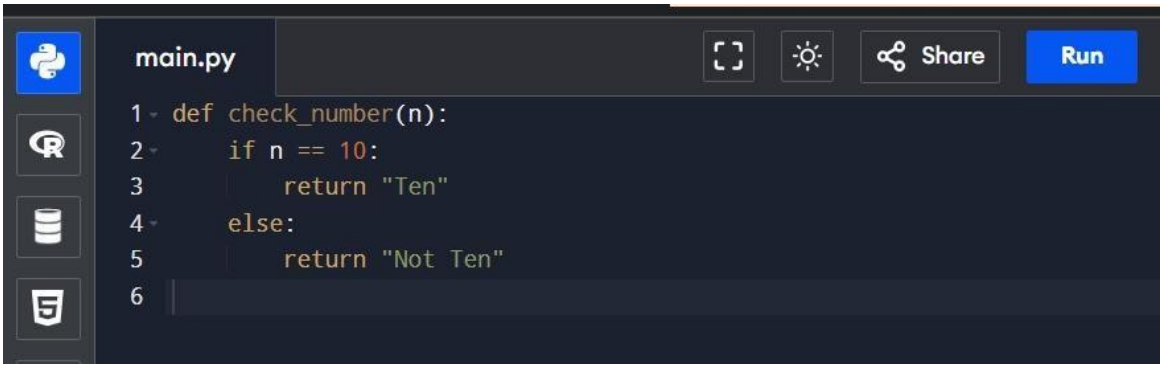
Corrected Code:

```
def check_number(n):
    if n == 10:
        return "Ten"
    else:
        return "Not Ten"
```

Assert Test Cases:

```
assert check_number(10) == "Ten"
assert check_number(5) == "Not Ten"
assert check_number(0) == "Not Ten"
```

Output:
All test cases passed



Task 3: Runtime Error – File Not Found

Corrected Code:

```
def read_file(filename):
    try:
        with open(filename, 'r') as f:
```

```
        return f.read()
    except FileNotFoundError:
        return "Error: File not found"
    except OSError:
        return "Error: Invalid file path"
```

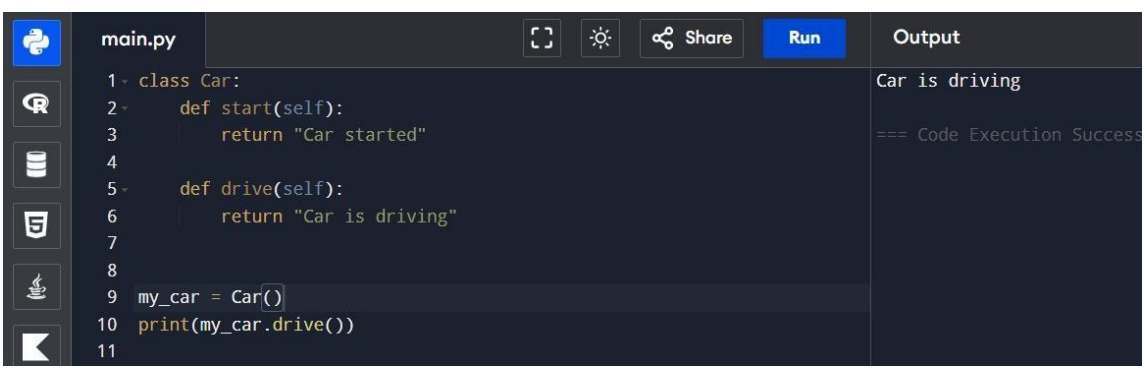
Output:
Error: File not found

Task 4: Calling a Non-Existent Method

Corrected Code:

```
class Car:
    def start(self):
        return "Car started"
    def drive(self):
        return "Car is driving"
```

Output:
Car is driving



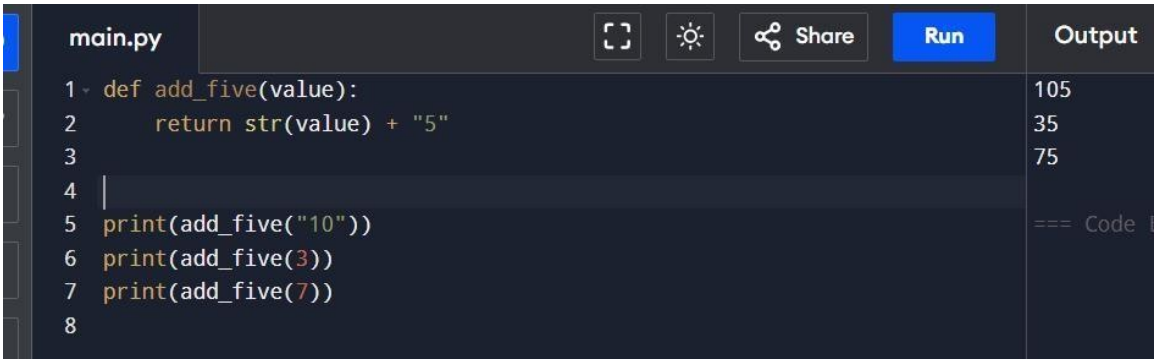
Task 5: TypeError – Mixing Strings and Integers

Solution 1:

```
def add_five(value):
   return int(value) + 5
```

Solution 2:

```
def add_five(value):
   return str(value) + "5"
```



Conclusion

This lab demonstrated how AI-assisted debugging helps identify errors, explain bugs clearly, and suggest safe fixes.